

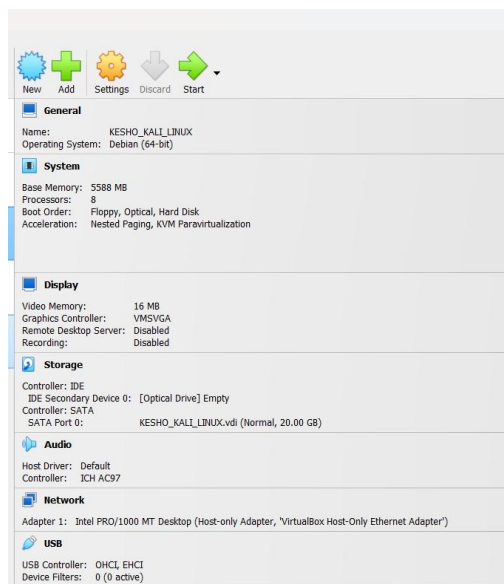
Ex: 2b Creating a VM and adding Storage, Memory, and Network to a VM**Date: 7/9/2026**

Aim:

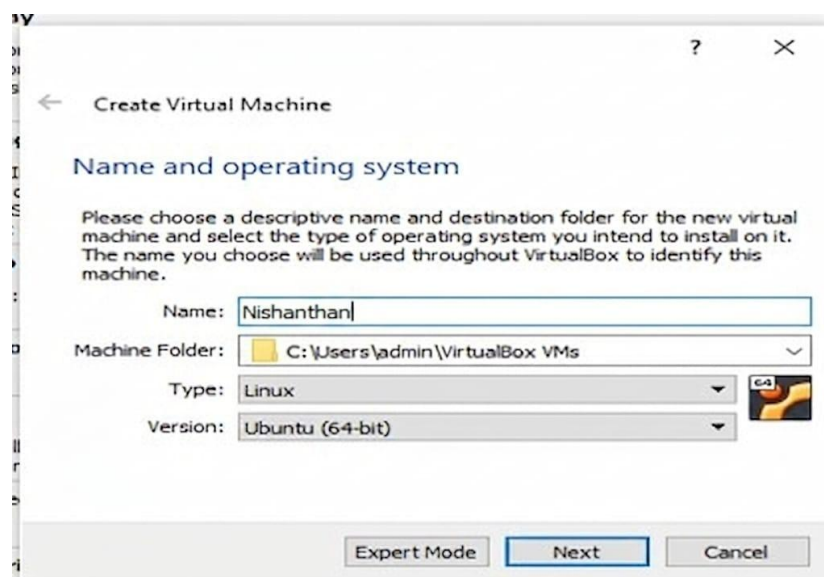
To create a vm and adding storage ,memory and network to a Vm in the oracle virtual box

Steps:

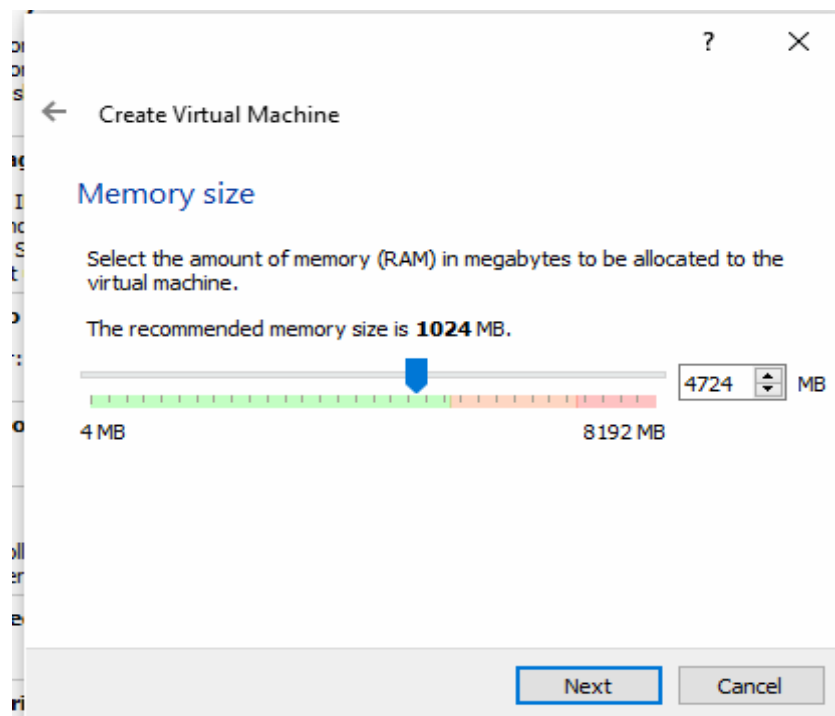
- 1) Open Oracle VM VirtualBox Manager and Click the "New" button



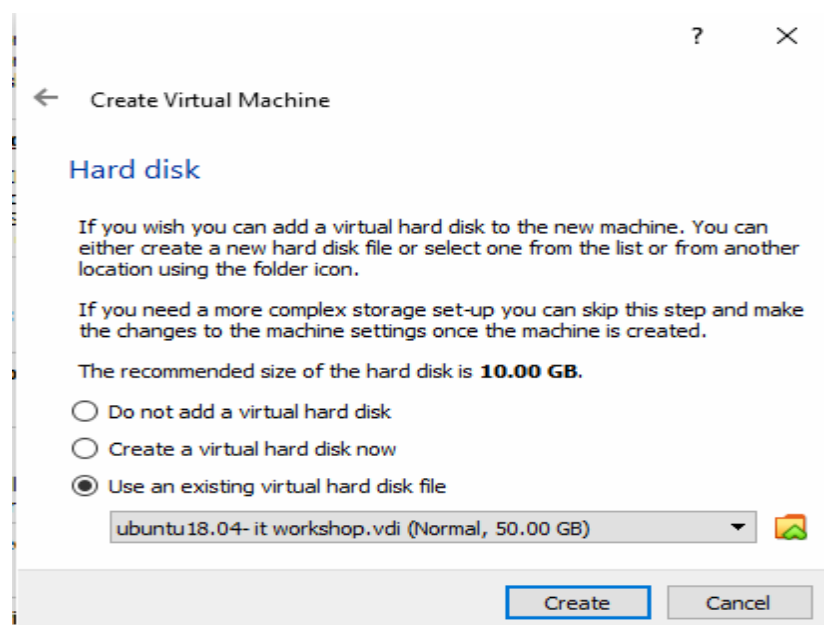
- 2) Fill the name: (KESHO_2A) Type: Linux Version: Ubuntu (64-bit) or your chosen OS and Click Next



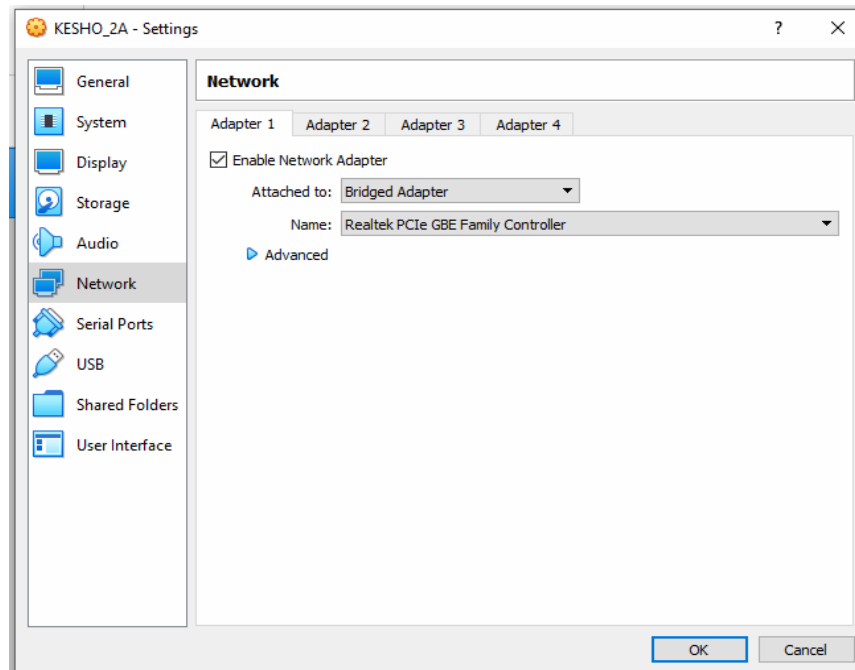
3) Allocate Memory (RAM) .Set the memory size 4gb has been set below Don't exceed 50% of your physical system RAM Click Next



4) Click use an existing virtual hard disk. Click Next Dynamically allocated Click Next Set disk size (e.g., 20 GB)Click Create



5) Configure Network Adapter Go to Settings Network Ensure Adapter 1 is: Enabled Attached to: Bridged Adapter (for internet + LAN access), or Host-Only Adapter (for communication with host only) Click OK Then new vm has been created



Execution of java program to test the execution time between vm (Ubunttu) and host(Windows)

1)To find the sum of digits between 1 to 1000000000

Code:

```

Activities Text Editor Thu 15:23
checker.java
~/23IT099
Save

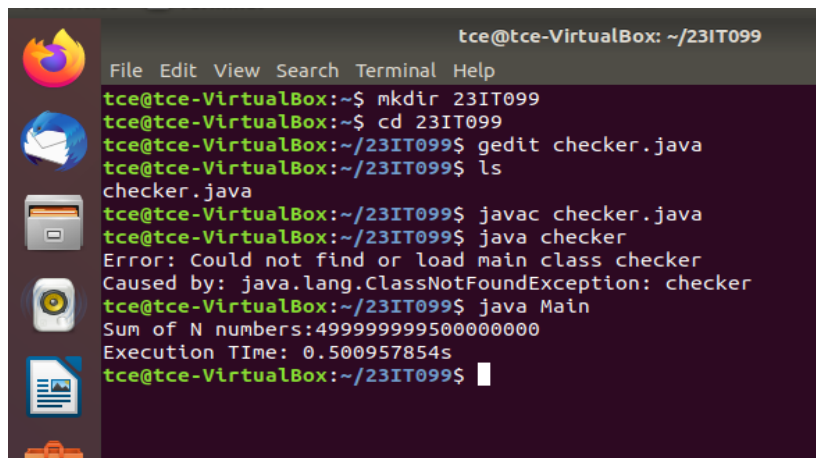
import java.util.*;

class Main
{
    public static void main(String args[])
    {
        long st = System.nanoTime();
        long nish = 0 ;
        for(int i=0;i<1000000000;i++)
        {
            nish+=i;
        }
        long et = System.nanoTime();
        double timeTaken = (et-st)/1000000000.0;
        System.out.println("Sum of N numbers:"+nish);
        System.out.println("Execution Time: " +timeTaken+"s");
    }
}

Java Tab Width: 8 Ln 18, Col 2 INS

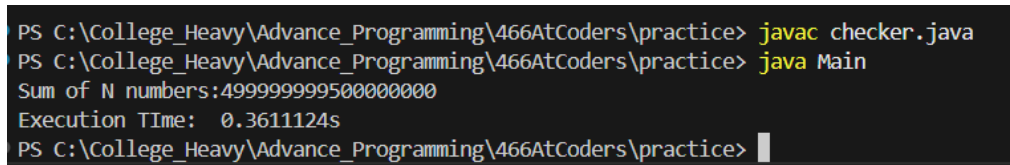
```

Execution time in Linux



```
tce@tce-VirtualBox: ~/23IT099
File Edit View Search Terminal Help
tce@tce-VirtualBox:~$ mkdir 23IT099
tce@tce-VirtualBox:~$ cd 23IT099
tce@tce-VirtualBox:~/23IT099$ gedit checker.java
tce@tce-VirtualBox:~/23IT099$ ls
checker.java
tce@tce-VirtualBox:~/23IT099$ javac checker.java
tce@tce-VirtualBox:~/23IT099$ java checker
Error: Could not find or load main class checker
Caused by: java.lang.ClassNotFoundException: checker
tce@tce-VirtualBox:~/23IT099$ java Main
Sum of N numbers:499999999500000000
Execution TIME: 0.500957854s
tce@tce-VirtualBox:~/23IT099$
```

Execution time in Windows



```
PS C:\College_Heavy\Advance_Programming\466AtCoders\practice> javac checker.java
PS C:\College_Heavy\Advance_Programming\466AtCoders\practice> java Main
Sum of N numbers:499999999500000000
Execution TIME: 0.3611124s
PS C:\College_Heavy\Advance_Programming\466AtCoders\practice>
```

From the above the execution time in **linux is 0.50 s** where the time taken by the **windows is 0.36** this clearly tells that the windows host is faster than the linux vm

Execution of C program to test the execution time between vm (Ubunttu) and host(Windows)

2) To find the sum of digits between 1 to 1000000000

Code:

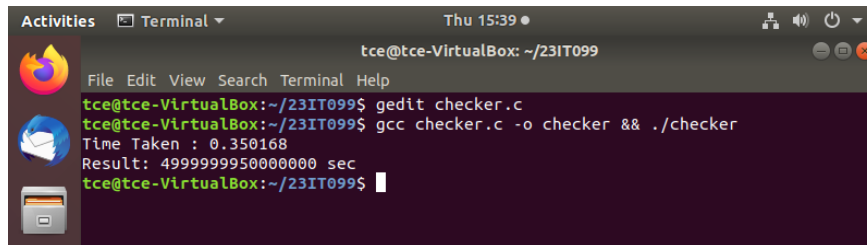
```
#include<stdio.h>
#include<time.h>

int main()
{
    clock_t start = clock();

    long result = 0;
    for(int i=0;i<1000000000;i++) result+=i;

    clock_t end = clock();
    double timeTaken = (double) (end-start)/CLOCKS_PER_SEC;
    printf("Time Taken : %lf \n",timeTaken);
    printf("Result: %ld sec\n",result);
}
```

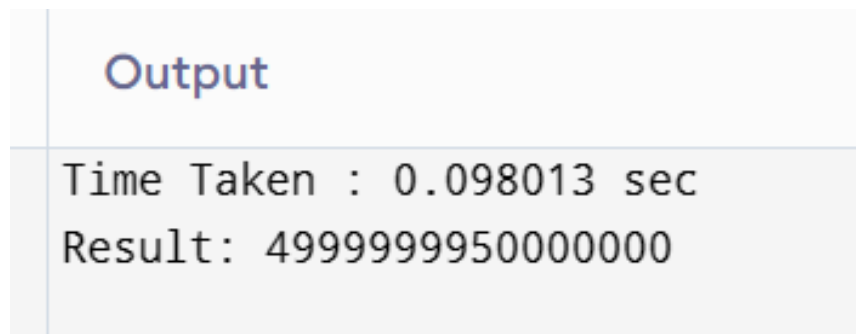
Execution time in Linux



The screenshot shows a terminal window titled 'Terminal' with the user 'tce' at host 'tce-VirtualBox' in directory '~/23IT099'. The terminal output is as follows:

```
tce@tce-VirtualBox: ~/23IT099
File Edit View Search Terminal Help
tce@tce-VirtualBox:~/23IT099$ gedit checker.c
tce@tce-VirtualBox:~/23IT099$ gcc checker.c -o checker && ./checker
Time Taken : 0.350168
Result: 4999999950000000 sec
tce@tce-VirtualBox:~/23IT099$
```

Execution time in Windows



The screenshot shows a window titled 'Output' with the following text:

```
Time Taken : 0.098013 sec
Result: 4999999950000000
```

From the above the execution time in **linux is 0.35 s** where the time taken by the **windows is 0.09** this clearly tells that the windows host is faster than the linux vm

RESULT:

Thus the Creation of VM and adding Storage, Memory, and Network to a VM has been created successfully and output has been verified.

